

DBMS PLACEMENT PREPARATION NOTES

Modules 3–5 • ER Model • Keys • Relational Model & Constraints

Goal: Short, placement-focused notes for MCQs, technical rounds, and DBMS interviews.

Efftronics focus: DBMS is explicitly listed in the Software Engineer assessment areas.

MODULE 3 — ENTITY RELATIONSHIP (ER) MODEL

ER Model: A conceptual model used to represent entities, attributes, and relationships.

1. Entity

Entity: A real-world object that can be uniquely identified. **Example:** Student, Employee.

2. Entity Set

Entity Set: Collection of similar entities.

3. Attribute

Attribute: A property that describes an entity.

4. Types of Attributes

- **Simple:** Cannot be divided further. Example: Age.
- **Composite:** Can be divided. Example: Name → First Name + Last Name.
- **Single-valued:** One value. Example: Date of Birth.
- **Multi-valued:** Multiple values. Example: Phone Numbers.
- **Derived:** Calculated from another attribute. Example: Age from Date of Birth.
- **Stored:** Actually stored in the database. Example: Date of Birth.

5. Relationship

Relationship: Association between entities. **Example:** Student enrolls in Course.

6. Degree of Relationship

- **Unary:** Same entity set. **Binary:** Two entity sets. **Ternary:** Three entity sets.

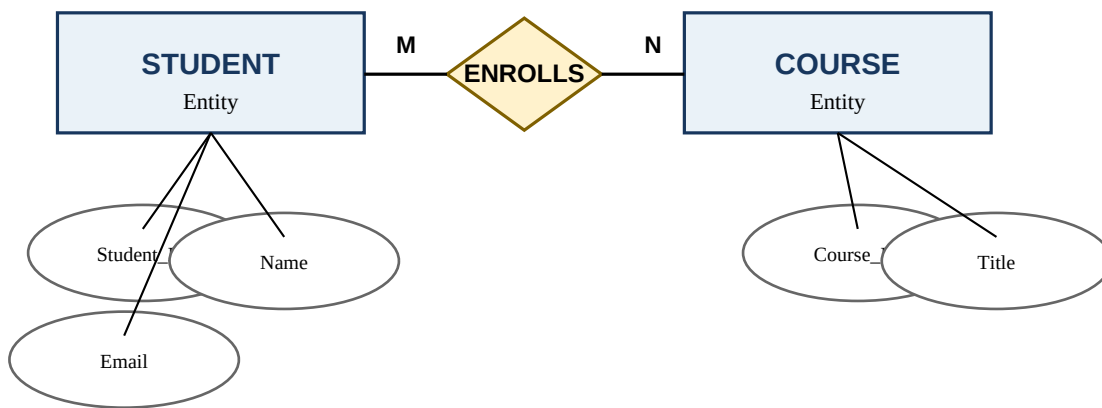
7. Cardinality

Cardinality: Number of entities participating in a relationship.

- **1:1** One-to-one **1:N** One-to-many **M:N** Many-to-many

8. Participation

- **Total:** Every entity must participate. **Partial:** Participation is optional.



Example ER diagram: Student M:N Course relationship

9. Strong vs Weak Entity

Strong Entity	Weak Entity
Has its own primary key.	Does not have a complete key of its own.
Exists independently.	Depends on a strong entity.

10. ER Diagram Symbols

Symbol	Meaning
Rectangle	Entity
Ellipse	Attribute
Diamond	Relationship
Double rectangle	Weak entity
Double ellipse	Multi-valued attribute
Dashed ellipse	Derived attribute
Underlined attribute	Key attribute

MODULE 4 — KEYS

Key: Attribute or set of attributes used to uniquely identify a record.

1. Super Key

Super Key: Any attribute set that uniquely identifies a row.

2. Candidate Key

Candidate Key: Minimal super key.

3. Primary Key

Primary Key: Candidate key selected to uniquely identify rows.

4. Alternate Key

Alternate Key: Candidate key not selected as primary key.

5. Foreign Key

Foreign Key: Attribute that references a key in another table.

6. Composite Key

Composite Key: Key made of two or more attributes.

7. Unique Key

Unique Key: Constraint that prevents duplicate values in a column/set of columns.

Key	Quick Definition
Super Key	Unique; may contain extra attributes.
Candidate Key	Minimal super key.
Primary Key	Selected candidate key.
Alternate Key	Candidate key not selected.
Foreign Key	References a key in another table.
Composite Key	Uses multiple attributes.

■ Placement Traps

- Every primary key is a candidate key, but not every candidate key is primary.
- Every candidate key is a super key, but not every super key is a candidate key.
- A foreign key can contain duplicate values.
- A composite key contains more than one attribute.

MODULE 5 — RELATIONAL MODEL & CONSTRAINTS

1. Relational Model

Relational Model: Represents data using tables called relations.

Term	Meaning
Relation	Table
Tuple	Row
Attribute	Column
Domain	Set of allowed values
Degree	Number of attributes/columns
Cardinality	Number of tuples/rows

2. Integrity Constraints

Integrity Constraint: Rule that maintains correctness and consistency of data.

Constraint	Meaning
Domain Constraint	Values must belong to the allowed domain.
Key Constraint	Key values must uniquely identify rows.
Entity Integrity	Primary key cannot be NULL.
Referential Integrity	Foreign key must reference a valid key value or be NULL where allowed.

3. SQL Constraints

Constraint	Purpose
PRIMARY KEY	Unique + NOT NULL identifier.
FOREIGN KEY	Maintains relationship between tables.
UNIQUE	Prevents duplicate values.
NOT NULL	Prevents NULL values.
CHECK	Enforces a condition.
DEFAULT	Provides a default value.

4. Primary Key vs UNIQUE

Primary Key	UNIQUE
Uniquely identifies rows.	Prevents duplicate values.
Cannot be NULL.	NULL handling depends on DBMS.
Normally one primary-key constraint per table.	A table can have multiple UNIQUE constraints.

5. Foreign Key Example

Department(Dept_ID, Dept_Name)

Employee(Emp_ID, Name, Dept_ID)

Employee.Dept_ID can reference Department.Dept_ID.

6. Placement Questions

- **Q:** What is a tuple? **A:** A row in a relation.
- **Q:** What is degree? **A:** Number of attributes/columns.

- **Q:** What is cardinality? **A:** Number of rows/tuples.
- **Q:** What is the minimal super key? **A:** Candidate key.
- **Q:** Which key connects tables? **A:** Foreign key.
- **Q:** Can a foreign key have duplicates? **A:** Yes.
- **Q:** Can a primary key be NULL? **A:** No.
- **Q:** Which constraint checks a condition? **A:** CHECK.

■ FINAL 1-MINUTE REVISION

Concept	Remember
Entity	Real-world object
Attribute	Property of entity
Relationship	Association between entities
Cardinality	1:1 / 1:N / M:N
Super Key	Unique; may have extra attributes
Candidate Key	Minimal super key
Primary Key	Selected candidate key
Foreign Key	References another table's key
Tuple	Row
Attribute	Column
Degree	Number of columns
Cardinality (relation)	Number of rows
Entity Integrity	Primary key $\neq$ NULL
Referential Integrity	Foreign key references valid key

Study method: First memorize the one-line definitions. Then redraw the ER diagram once and solve the placement questions without looking at the answers.